RAG-Based AI Teaching Assistant

Synopsis submitted in the partial fulfillment of the requirements
for the award of the degree of

BACHELOR OF TECHNOLOGY
in

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by

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CANDIDATE'S DECLARATION

We hereby declare that the work which is being presented here entitled, ***" RAG-Based AI Teaching Assistant"*** by **Pragya verma, Ankush kr poddar, Lalbabu yadav** in partial fulfillment of requirements for the award of the degree of B.Tech. in Information Technology submitted in the Department of Information Technology, PIET, Panipat affiliated to Kurukshetra University, is an authentic record of our own work under the supervision of **Ms.Urvinder kaur.** The matter presented in this synopsis has not been submitted by us in full or in part to any other University / Institute for the award of B. Tech degree.

Signature of the Students Date:

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This is to certify that the above statement made by the candidate is correct to the best of our knowledge.

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Abstract

This project proposes the development of a **RAG-Based AI Teaching Assistant** that provides accurate, syllabus-focused, and context-aware academic support. By integrating a pre-trained Large Language Model with approved study materials such as lecture notes, PDFs, and textbooks, the system retrieves relevant content before generating responses, ensuring high accuracy and reduced hallucination. Using techniques like embeddings and vector databases, the platform delivers exam-oriented, reliable answers without requiring model retraining. The system is built using open-source tools, making it cost-effective, scalable, and suitable for real-world educational use.

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1. Introduction of the Problem and Objectives

General Introduction

With the rapid growth of digital learning platforms and AI-based tools, students increasingly rely on online systems for academic support. However, most existing AI assistants provide **general-purpose answers** that are not aligned with specific academic syllabi, leading to confusion, irrelevant explanations, and reduced exam preparedness. These systems often generate responses based on broad internet knowledge, which may result in **out-of-syllabus content, hallucinated answers, and lack of academic reliability**.

This project aims to address these challenges by developing a **RAG-Based AI Teaching Assistant** that integrates **Retrieval-Augmented Generation (RAG)** with approved academic content such as lecture notes, textbooks, PDFs, and syllabi. By combining modern techniques with vector-based semantic search and large language models, the system ensures **high accuracy, reduced hallucination, and syllabus-aligned responses**, offering reliable academic support to students and educators.

Specific Objectives

- To design and develop an AI-based teaching assistant using Retrieval-Augmented Generation (RAG).
- To create a structured academic knowledge base using approved lecture notes, PDFs, and syllabi.
- To implement semantic search using embeddings and vector databases for accurate content retrieval.
- To generate context-aware, syllabus-focused, and exam-oriented answers using a large language model.
- To reduce hallucinated and incorrect responses through document-grounded answer generation.
- To provide a scalable, cost-effective, and easy-to-update learning support platform.

Justification of Objectives

This project enhances traditional AI systems by grounding responses in authorized academic material, improving accuracy and reliability. The use of open-source tools makes the system cost-effective and scalable, making it suitable for real-world educational environments.

Strengths and Limitations

Strengths:

- Syllabus-aware and accurate AI-based learning support
- Reduced hallucination using document-grounded responses
- Scalable and modular AI architecture
- Cost-effective system using open-source tools

Limitations:

- Performance depends on the quality of uploaded academic content.
- High user load may require better computational resources.
- Limited offline functionality

2. Literature review

- **AI-Based Educational Assistants (2022–2024)**

  Recent studies highlight the increasing use of AI-powered teaching assistants to support student learning. Traditional AI systems generate generic responses and often lack syllabus alignment, which can lead to irrelevant or incorrect answers. Research shows that integrating **document-based retrieval with language models** significantly improves answer accuracy, contextual relevance, and academic reliability. These systems help in providing personalized, exam-oriented, and trustworthy learning support[1][2][3].

- **Retrieval-Augmented Generation (RAG) Systems (2023–2025)**— Recent advancements in Retrieval-Augmented Generation (RAG) show that combining semantic search with large language models significantly improves answer accuracy and reduces hallucination. By retrieving relevant documents before generating responses, RAG-based systems outperform standalone LLMs in domain-specific tasks [3][4].

- **Semantic Search and Vector Databases (2022–2024)**

  Recent advancements in Retrieval-Augmented Generation (RAG) show that combining semantic search with large language models significantly improves answer accuracy and reduces hallucination.This approach enables scalable, cost-effective, and continuously updatable AI solutions, making it highly suitable for educational applications[4][5][6].

- **AI in Personalized Learning Systems (2022–2024)**

  Studies in intelligent tutoring systems emphasize the role of AI in enhancing personalized learning experiences. AI-based platforms improve concept clarity, adaptive learning, and student engagement. By grounding responses in curated academic content, these systems provide reliable educational support while reducing dependency on unverified online sources[7][8].

3. Methodology and Workflow

Tentative Strategy

1. **Requirement Analysis**

   - Identify core features such as syllabus-based question answering, document retrieval, and intelligent response generation.
   - analyze student needs for accurate, exam-oriented, and learning support.

2. **System Design**

   - Architecture: RAG-based system using LLM, vector database, and retrieval pipeline
   - AI Layer: Pre-trained LLM with embeddings and semantic search
   - Database Design: Vector database for storing document embeddings and metadata

3. **Implementation**

   - **Frontend:** User-friendly web interface for asking academic questions and viewing AI-generated responses.

     Simple and intuitive dashboard for easy interaction
   - **Backend:** Integration of RAG pipeline using embeddings, vector database, and LLm

     API-based system for document processing, semantic retrieval, and intelligent answer generation
   - **Database:** For storing document embeddings and enabling fast semantic search

     For storing user queries, chat history, system logs, and metadata

4. **Testing**

   - Validation of question-answering accuracy
   - API testing for document ingestion, retrieval, and response generation

5. **Deployment & Evaluation**

   - Deployment using **free cloud hosting platforms.**
   - Continuous **performance monitoring and feedback collection.**

4. Hardware and Software Requirements

Hardware Requirements:

- Processor: Intel Core i3 / i5 or higher
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: 100 GB HDD or 256 GB SSD
- Peripherals: Keyboard, Mouse, Monitor

Software Requirements:

- Operating System: Windows 10 / Mac os
- Development Environment: Visual Studio Code
- Programming Languages: Python, JavaScript
- Database: Vector Database (FAISS / ChromaDB)
- Frameworks & Libraries: Flask / FastAPI, React
- Additional Tools: GitHub (version control), Postman (API testing)

5. Project Title Map with Engineering Program Outcomes and Program/Department Specific Outcomes

Project Title	**Engineering Program Outcomes (POs)**	**Department Specific Outcomes (DSOs)**
RAG-Based AI Teaching Assistant for Syllabus.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12	PSO1, PSO2

6. Project Categorization

- **Product-based** – Development of a functional AI-powered teaching assistant platform
- **Application-based** – Solves real-world academic learning and doubt-clearing problems
- **Research-based** – Applies advanced AI techniques like Retrieval-Augmented Generation (RAG), semantic search, and NLP.

Project Categorization: Application-based

7. Project Map with SDG Goals

- **SDG 4: Quality Education** – Provides an intelligent, syllabus-aware, and accessible AI-based learning assistant that supports students
- **SDG 9: Industry, Innovation, and Infrastructure** – Promotes innovation in educational technology by developing a scalable, intelligent AI system using modern digital infrastructure

8. Conclusions

The **RAG-Based AI Teaching Assistant** project demonstrates how artificial intelligence can significantly enhance traditional learning systems by providing accurate, syllabus-aware, and intelligent academic support. By integrating Retrieval-Augmented Generation with modern AI techniques, the system delivers a reliable, scalable, and cost-effective platform for personalized learning. The project emphasizes accessibility, innovation, and practical implementation, making it suitable for academic institutions and real-world educational environments while contributing to the advancement of intelligent learning systems.

9. Specific Contributions of Team Members

Team Member	Contribution
Member 1	Frontend Development (React, UI/UX)
Member 2	Backend & Database (Node.js, Express, VectorDB)
Member 3	Api Integration & Backend development

10. Key References

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